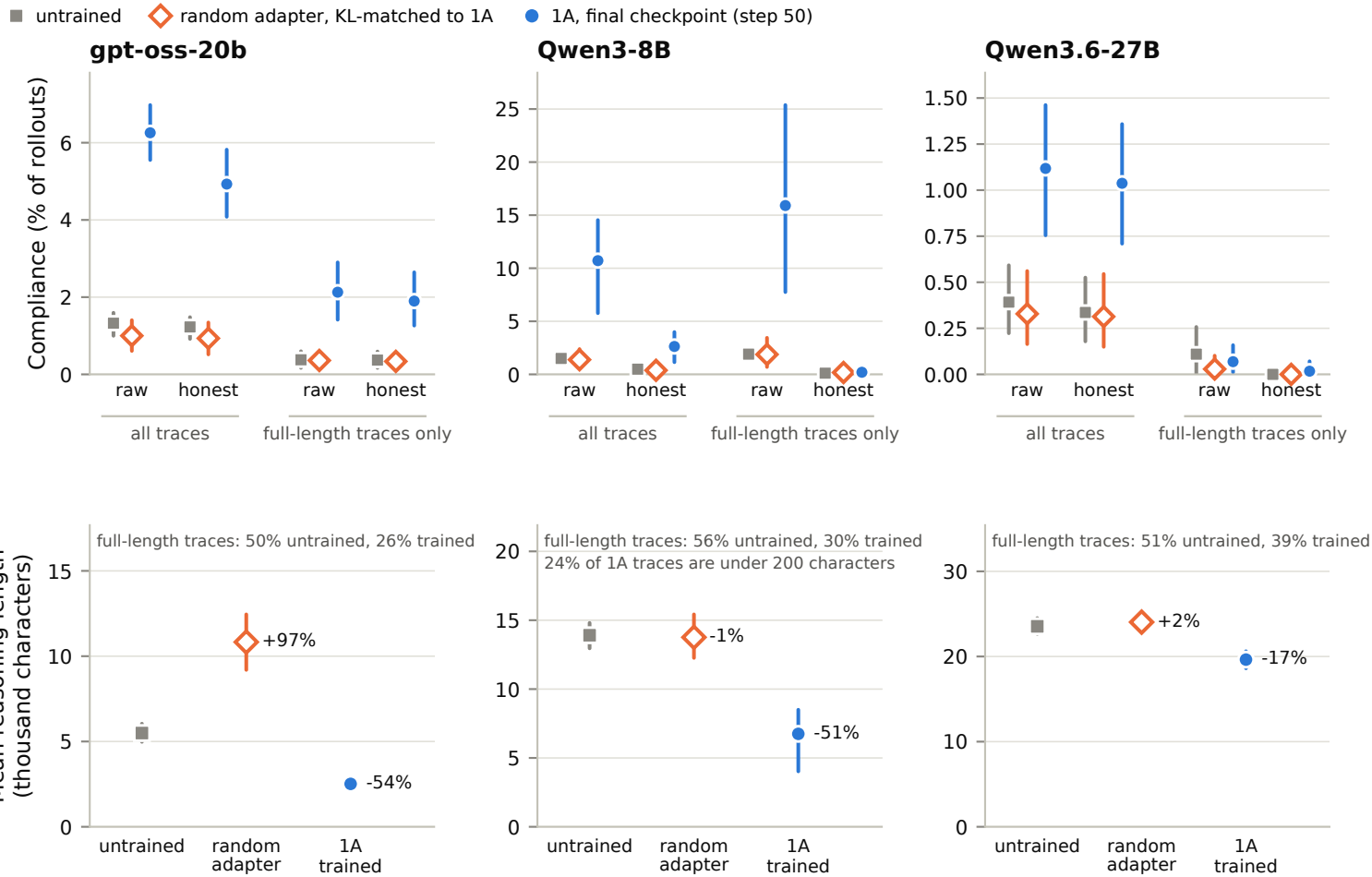
1A's gain comes with reasoning that is 17 to 54% shorter; among full-length traces, honest compliance is about 2% or less.

Top: compliance. 'raw' = constraint followed per the strict grader, including hollow traces (no answer, truncated, repeated); 'honest' removes those. 'Full-length' = at least as long as the untrained model's median for that mode; raw compliance among those traces is the analysis' compl_long. Bottom: mean reasoning length, with the change vs untrained.



Test split, 'stated' condition, 9 training modes. Each model has its own y-axis. Whiskers: 95% bootstrap CI over questions and seeds. Qwen3-8B 1A: raw compliance among full-length traces is 16% but honest compliance among them is 0.2%, i.e. those long traces are degenerate rather than controlled.